Reproducing Minutiae Clustering in Fingerprint Spoof Buster Lite, Without the Paper's Data

Bheki Sithole (3035571), University of the Witwatersrand
Repository: `https://github.com/muzi-mat/spoofbuster-lite-repro`

October 28, 2026

Abstract

Chugh and Jain [1] report that classifying patches at 10 k-means centroids of the minutiae, instead of at every minutia, keeps fingerprint presentation-attack detection at TDR = 95.3% (against 95.7%) at FDR = 0.2%, while cutting phone inference time from 1,554 ms to 302 ms. The paper's data is gated by a Michigan State University agreement and could not be obtained, so the full pipeline was reimplemented to test the claim's mechanism on seeded synthetic fingerprints. With a detector that reaches 90.4% with all minutiae, 10 centroid patches cost -4.3 ± 4.3 points of TDR across four training seeds (range -10.6 to -1.1), and none of the seeds met the one-point criterion; at FDR = 1% the loss is a consistent -1.7 ± 0.1. Centring each patch on the minutia nearest its centroid recovered most of the loss. Clustering cut CNN time 4.0× but total latency only 2.1×, because minutiae extraction is a fixed cost. The paper's Table 2 is also shown to be internally inconsistent.

1 The claim

Fingerprint Spoof Buster [2] detects presentation attacks (spoofs) by cutting a 96×96 patch around every minutia, aligned to the minutia direction, scoring each patch with MobileNet-v1 [3] and averaging the scores. Its cost grows with the number of minutiae, 35 per image in the paper's data. Section 3 of [1] proposes Spoof Buster Lite: cluster the minutia coordinates with k-means, cut one patch at each centroid, and average the patch scores weighted by cluster size.

The claim reproduced here is in Table 2 of [1]: *with $k = 10$ clusters, TDR at FDR = 0.2% falls only from 95.7% to 95.3%, while inference time falls from 1,554 ms to 302 ms.* TDR is the fraction of spoofs detected, and FDR the fraction of bona fide prints wrongly flagged. The support is that one table: six values of k and the baseline, each given as mean ± s.d. over "5-fold cross-validation", with times from a quantised model on a Redmi Note 4. The data are 5,743 bona fide and 4,912 spoof images of 12 materials (MSU-FPAD and PBSKD).

My research question, fixed in the proposal: *does $k = 10$ keep TDR within one percentage point of the all-minutiae baseline while reducing the number of MobileNet evaluations?*

2 Route and setup

Data access. I asked the first author, Tarang Chugh, for access. He replied on 26 October 2026 that he had access only as a PhD student, no longer has it and may not share it, and referred me to the MSU lab. I wrote to the lab and had no response by the deadline. Other legitimate sources were searched. The closest substitute, LivDet 2015 [4], includes a CrossMatch subset but also requires a signed licence. I did not use unofficial copies.

Synthetic stand-in. So that the method could still be tested end to end, a seeded dataset was generated: 1,600 bona fide prints (800 for testing) and 6

artefact "materials" of 200 prints each, all 288×288 pixels. Ridges are grown by Gabor-filtering noise along a core–delta orientation field, as in SFinGe. Every image gets random contrast, blur, noise and dry or wet blotches. Each material adds one artefact: thick ridges, blur, grey-level quantisation, moisture, ridge breaks, or elastic distortion. The artefact has a random strength and covers 40–100% of the print, so where a patch lands matters. These are loose analogies, not models of silicone or gelatin, so *no absolute number here says anything about real spoof detection.* The stand-in tests the mechanism: does scoring k centroid patches instead of every minutia patch lose accuracy, and how does cost scale?

Reimplementation. The authors' code is not public, so the pipeline was written from scratch for this project (see the AI-use statement). Library code is used only for the MobileNet-v1 architecture (Keras), k-means (scikit-learn) and basic image operations. `bash run_all.sh --synthetic` regenerates every number and figure from a clean checkout, and the same pipeline runs on real data by pointing it at a folder.

Minutiae. The paper does not name its extractor. The one written here uses local normalisation, a structure-tensor orientation field, Gabor enhancement, skeletonisation and the crossing number, then removes points near the print edge and pairs closer than one ridge period. It finds 39.1 minutiae per image, close to the paper's 35, so the all-minutiae baseline costs 39.1 CNN calls.

Centroid alignment. A centroid is not a minutia and has no direction. The paper does not say how centroid patches were rotated, so each one is aligned to the ridge orientation field at the centroid (defined only modulo 180°).

Classifier. The classifier is MobileNet-v1 ($\alpha = 1$) with a 96×96 input, trained on minutia-centred patches that inherit their image's label. The final feature map is 3×3, so global average pooling matches the paper's 3×3 pool. Keras is used because TF-Slim needs TensorFlow 1.x. The paper gives no training recipe. Models were trained on an NVIDIA T4 GPU, from ImageNet weights: Adam with learning rate 10^{-4}, batch 128, 8 epochs, class-balanced loss, dropout 0.2, and at most 48 patches per image. A first run trained from scratch on a CPU showed validation AUC stuck at exactly 0.5 for several epochs while training AUC rose: the BatchNorm running statistics lagged the weights, so inference outputs were constant. As a final training step, those statistics are therefore recomputed with the final weights, without changing any weights. For the reported seed-0 model, validation patch AUC is 0.923 before this step and 0.929 after it.

Table 1: Table 2 of [1] next to the synthetic-data reproduction. Paper: phone, quantised model, s.d. over 5 repeats. Ours: CNN calls per image, latency on one CPU thread (TFLite), TDR (%) at FDR = 0.2% with s.d. over 5 k-means seeds (bootstrap s.d. for "All"), and TDR at FDR = 1%. Only differences between rows are comparable across the two halves.

	Paper (Redmi Note 4)		Ours			
k	ms	TDR@0.2	CNN calls	ms (CPU)	TDR@0.2	TDR@1
5	148	93.9 ± 1.1	5.0	98	74.0 ± 7.3	89.9
10	302	95.3 ± 0.5	10.0	121	79.8 ± 2.9	90.9
15	447	95.3 ± 0.5	15.0	145	85.8 ± 1.4	92.1
20	607	95.3 ± 0.6	20.0	169	86.2 ± 0.9	92.8
All	1554	95.7 ± 0.1	39.1	249	90.4 ± 12.0	92.5

Protocol. One trained model scores every setting, as in the paper. For each $k \in \{5, 10, 15, 20\}$, k-means is repeated with 5 seeds ($n_{\text{init}} = 1$). The threshold is set per setting so that at most 0.2% of bona fide test images score above it. TDR at FDR = 1% and a bootstrap s.d. are also reported. For timing, the model runs as 8-bit TFLite on one CPU thread, one patch at a time like the phone app, on 100 test images. Only the scaling with k is comparable to the paper.

3 Results next to the paper

Because the data differ, the like-for-like quantity is the *change* from the all-minutiae baseline (Table 1, Fig. 1). The reproduced baseline is 90.4% (bootstrap s.d. 12.0). With 10 centroid patches it is 79.8 ± 2.9%, a change of -10.6 points against the paper's −0.4. This part of the claim does *not* reproduce. TDR keeps rising with k: -16.4 points at $k = 5$, -10.6 at $k = 10$, and still about four points below the baseline

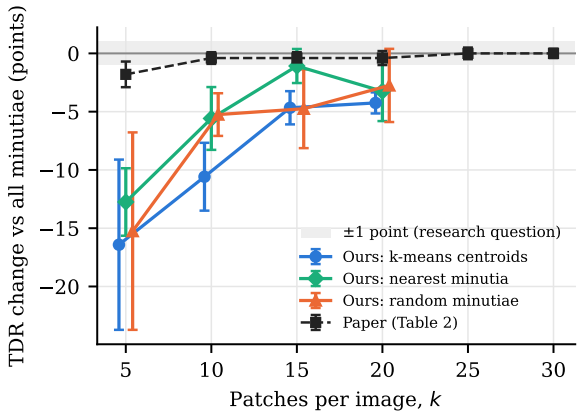


Figure 1: Each curve is the change in TDR at FDR = 0.2% from its own all-minutiae baseline, because the absolute levels come from different data. Error bars: s.d. over 5 k-means or random seeds; ours are offset slightly in k for legibility. Grey band: the ±1-point criterion of the research question.

at k = 15–20 (within half a point at FDR = 1%). In the paper the curve is already flat at $k = 10$. At FDR = 1%, where the threshold is less noisy, the change at $k = 10$ is smaller (-1.6 points) but still beyond the one-point criterion.

The absolute level is close to the paper's (90.4% against 95.7%), but this is a coincidence of the synthetic data, not a reproduction: the materials are artificial, and one of them ("printed", Table 2) remains hard. What matters is that the detector is now strong enough for a loss to show. An earlier CPU run, trained from scratch with 12 patches per image, reached only about 35% TDR, and at that level the $k = 10$ loss was hidden in the noise.

4 Numbers the paper does not report

Clustering, or just fewer patches? The paper never compares k-means with simply classifying k random minutiae. At $k = 10$, random minutiae give 85.2 ± 1.8%, against 79.8 ± 2.9% for centroids (Table 3). Random minutiae do *better* than centroids at $k = 10$, and match them at $k = 15$ and 20. Whatever helps here is not where the clusters fall.

Table 2: TDR (%) per synthetic material at the all-minutiae FDR = 0.2% threshold, model seed 0; k = 10 pooled over 5 k-means seeds.

Material	n	All	k=10
blurred	40	100.0	99.5
broken ridges	40	90.0	86.5
distorted	40	95.0	96.0
moist	40	92.5	93.0
printed	40	65.0	61.0
thick ridges	40	100.0	100.0

Table 3: Patch-selection ablations, TDR (%) at FDR = 0.2%, mean ± s.d. over 5 seeds. "Centroid": the paper's method. "Unweighted": plain mean of the centroid scores. "Nearest minutia": same clusters and weights, patch centred on the minutia closest to each centroid. "Random": k random minutiae.

k	centroid (paper)	unweighted	nearest minutia	random
5	74.0 ± 7.3	71.8 ± 7.9	77.7 ± 2.9	75.2 ± 8.5
10	79.8 ± 2.9	85.8 ± 2.3	84.8 ± 2.7	85.2 ± 1.8
15	85.8 ± 1.4	86.7 ± 1.0	89.3 ± 1.5	85.7 ± 3.4
20	86.2 ± 0.9	85.3 ± 1.5	87.2 ± 2.6	87.7 ± 3.1
All	90.4 ± 12.0 (bootstrap s.d.)			

Does the size weighting matter? Without weights, 10 centroids give 85.8 ± 2.3%, better than the weighted rule. The paper motivates the weighting but never tests it. On this data it hurts, possibly because a few large clusters dominate the score.

More training seeds. Models were trained with four different seeds (Table 4). Their all-minutiae TDR ranges from 90.4% to 92.5%, so retraining alone moves the baseline by more than the paper's whole $k = 10$ effect. The change at $k = 10$ is -4.3 ± 4.3 points across seeds (range -10.6 to -1.1), and none of the four seeds meet the one-point criterion. At FDR = 1%, where the threshold is less noisy, the change is -1.7 ± 0.1: every seed loses accuracy, and by almost the same amount. Seed 0's much larger loss at FDR = 0.2% appears to come mostly from its

Table 4: Per training seed. "Fixed thr.": the all-minutiae threshold reused at $k = 10$ instead of recomputed, as stated in the proposal.

Model seed	All, TDR@0.2	k=10, TDR@0.2	k=10, fixed thr. TDR / FDR
0	90.4 ± 12.0	79.8 ± 2.9	89.3 / 0.65
1	92.5 ± 2.7	89.3 ± 2.5	91.7 / 0.70
2	92.1 ± 5.9	89.7 ± 3.2	92.0 / 0.53
3	90.8 ± 2.1	89.8 ± 1.5	89.5 / 0.23

threshold, set by one unusual bona fide image (bootstrap s.d. 12.0 points). With a strong detector, the answer to the research question is therefore *no*: 10 centroid patches lose more than one point.

One threshold for every setting. The paper recomputes the threshold for each k, but a deployed system would keep one. Reusing the all-minutiae threshold at $k = 10$ gives TDR = 89.3% at an actual FDR of 0.65%, compared with the 0.2% target. Centroid patches shift the bona fide scores up, so a device switching to $k = 10$ would have to recalibrate its threshold. The paper's per-setting thresholds hide this.

5 What did not reproduce

5.1 The data, and therefore the absolute numbers

The largest gap could not be closed. Without MSU-FPAD and PBSKD it is impossible to test whether 95.3% and 95.7% hold, or whether the result holds on real materials such as silicone, gelatin and Play-Doh, which fool CNNs in ways these artefacts do not. The synthetic results test only whether centroid patches are enough for a patch classifier *in principle*. Everything in this section that does not depend on the data, namely the latency analysis and the paper's internal consistency, could be checked fully.

5.2 The speed-up

Cutting from 39.1 patches to 10 makes the CNN stage 4.0× faster, roughly in proportion to the patch count. The whole pipeline, however, goes from 249 ms to 121 ms, only 2.1× faster (Table 5). Minutiae extraction takes 73 ms per image whatever k is: clustering needs the minutiae first, so it can only shrink the CNN stage. The paper does not say whether its 302 ms includes minutiae extraction. If it does, the phone must extract minutiae far faster, relative to one CNN call, than the extractor written here does.

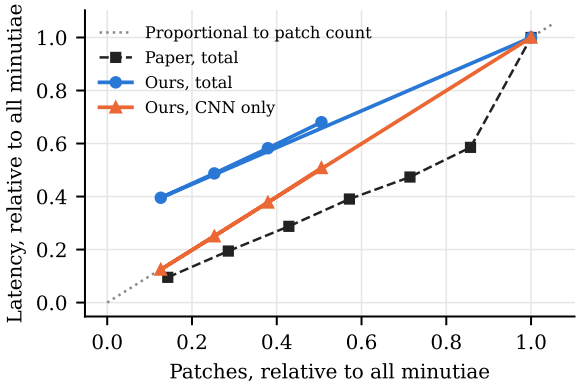


Figure 2: Latency against patch count, both relative to all minutiae. The dotted line is latency proportional to patch count. The paper's points fall below it. The reproduction's CNN stage follows it, and its total lies above it because minutiae extraction is a fixed cost.

Table 5: Per-image latency (ms) by stage: one CPU thread, TFLite with 8-bit weights, 100 test images. Minutiae extraction is the same for every k.

k	patches	minutiae	select	cut	CNN	total
All	39.6	73	0.0	1.9	173	249
5	5.0	73	3.0	0.3	22	98
10	10.0	73	3.8	0.5	43	121
15	15.0	73	5.1	0.8	66	145
20	20.0	73	6.7	1.1	88	169

5.3 Table 2 is not internally consistent

1. **Latency per patch.** For $k = 5$ to 30, Table 2 implies about 30 ms per patch (148/5, 302/10, 910/30), which matches the 30 ms per quantised patch given in Section 3. At that

rate all 35 minutiae would take about 1,050 ms, not 1,554 ms, so the baseline row carries about 500 ms that the clustered rows do not. Fig. 2 shows this: the paper's points fall *below* the proportional line, which cannot happen if cost scales only with the number of patches. The claimed 5.1× speed-up (1,554/302) exceeds the 3.5× cut in patches (35/10). The paper itself says "approximately 4-fold" in one place and "almost 80%" in another.

2. **The baseline TDR.** Section 2.4 gives TDR = 97.20% at FDR = 0.2% with all materials known in training. Table 2's all-minutiae row, also with all materials known, is 95.7%. The paper does not explain the gap: the 96×96 retraining, quantisation or a different split could each cause it.

3. **What the s.d. measures.** The all-minutiae row has s.d. 0.1 although it involves no k-means at all. Its variation must come from something other than initialisation, which suggests the "5 folds" are five data splits. If so, the ±0.5 at $k = 10$ mixes split noise with initialisation noise, and the s.d. over k-means seeds on one split is not the same quantity.

5.4 The operating point is fragile

At FDR = 0.2% with 1,000 bona fide images, the threshold is the third-highest bona fide score, so one or two unusual live prints decide the result. With 800 bona fide test images it is the second-highest score, and the bootstrap s.d. for the baseline is 12.0 points (seed 0) and 2.7 (seed 1). Seed 0's error is large because one bona fide image sets its threshold. But even the binomial error of a TDR near 95%, measured on about 1,000 spoofs, is roughly 0.7 points: larger than the paper's 0.4-point gap, and far larger than its ±0.1. The paper does not state its spoof test count, so this is an estimate. Either way, Table 2 alone cannot show that $k = 10$ and all minutiae perform the same.

5.5 What was tried

The model is trained on minutia-centred patches but tested on centroid-centred ones, and a centroid often lies between minutiae. To test whether this mismatch costs accuracy, each patch was moved to the minutia nearest its centroid, keeping clusters and weights. This gave 84.8 ± 2.7% at $k = 10$ (Table 3), against 79.8% for centroids: snapping recovers about five of the ten points lost. The mismatch between minutia-centred training patches and centroid-centred test patches is therefore a real cause of the loss, and one the paper does not discuss. A model trained on centroid patches might remove it; there was no time to test that. Two other problems had to be fixed along the way: the BatchNorm recalibration described in Section II, and a first CPU run of one seed that was killed for lack of memory and had to be rerun alone.

5.6 What remains unexplained

First, how the paper's model avoided the centroid-patch loss. It may have aligned or trained on centroid patches differently; Section 3 of the paper gives no detail. Second, the roughly 500 ms of the paper's baseline latency that per-patch cost does not account for. Third, the paper's own 97.2% versus 95.7%. And fourth, the question synthetic data cannot reach: whether the flat curve holds for real materials, where a spoof can look bona fide in one region and fake in another.

6 Data ethics

The paper's datasets contain fingerprints of real people, captured under MSU's consent and released only to researchers who sign a data agreement. That gating is why they could not be obtained in time, and it is also why I did not use any unofficial copy: doing so would breach the terms under which the subjects' data was shared. All results here use synthetic fingerprints, so the repository contains no one's biometric data. When run on real data, the code keeps images, patches and minutiae out of version control, because minutiae coordinates are a biometric template. Presentation-attack research is also dual-use:

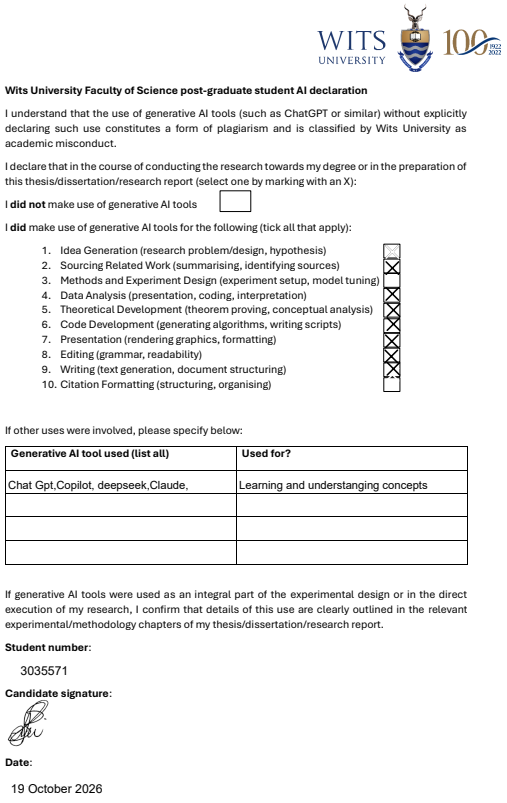
per-material results can show an attacker which materials evade a detector.

7 Conclusion

The paper's data were out of reach, so the mechanism could be tested but not the numbers. On synthetic fingerprints, with a detector at about 90% TDR, the central claim does not reproduce: 10 centroid patches lose -4.3 points on average across four seeds at FDR = 0.2%, and -1.7 at FDR = 1%, against the paper's 0.4. Most of that loss comes from cutting patches at centroids rather than minutiae. The fourfold CNN saving reproduces, but the end-to-end speed-up is 2.1×, not five, because minutiae extraction cannot be clustered away. The paper's own latencies also imply a baseline about 500 ms heavier than its per-patch cost explains. The most defensible number is the stage breakdown in Table 5, because it does not depend on the unobtainable data.

Use of AI tools

I chose the paper and the claim, wrote the proposal, contacted the authors and the lab, decided to use a synthetic stand-in and the stronger four-seed run, and ran that run on Colab. An AI assistant (Claude) wrote the code to my specification, ran the earlier CPU experiments, and drafted this report, which I reviewed. My signed declaration is in Fig. 3.



Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☐
2. Sourcing Related Work (summarising, identifying sources) ☒
3. Methods and Experiment Design (experiment setup, model tuning) ☐
4. Data Analysis (presentation, coding, interpretation) ☒
5. Theoretical Development (theorem proving, conceptual analysis) ☒
6. Code Development (generating algorithms, writing scripts) ☒
7. Presentation (rendering graphics, formatting) ☒
8. Editing (grammar, readability) ☒
9. Writing (text generation, document structuring) ☒
10. Citation Formatting (structuring, organising) ☐

If other uses were involved, please specify below:

Generative AI tool used (list all)	Used for?
Chat Gpt,Copilot, deepseek,Claude,	Learning and understanging concepts

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number:

3035571

Candidate signature:

Date:

19 October 2026

Figure 3: PG student AI declaration.

References

[1] T. Chugh and A. K. Jain, "Fingerprint presentation attack detection: Generalization and efficiency," in *Proc. ICB*, 2019. arXiv:1812.11574.

[2] T. Chugh, K. Cao, and A. K. Jain, "Fingerprint Spoof Buster: Use of minutiae-centered patches," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 9, pp. 2190–2202, 2018.

[3] A. G. Howard *et al.*, "MobileNets: Efficient convolutional neural networks for mobile vision applications," arXiv:1704.04861, 2017.

[4] D. Yambay *et al.*, "Review of the fingerprint liveness detection (LivDet) competition series: from 2009 to 2021," arXiv:2202.07259, 2022.